

AI Principles and Practice · COMP-0484

Ethics, Bias and Fairness

Week 3 · Reference Booklet

A plain-English guide: the ethics, the fairness metrics, and a worked audit

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How to use this booklet

This booklet is the long, friendly companion to the Week 3 lecture. Where the slides carry the argument and the lab puts it into practice, this document slows everything down: the full definitions, the arithmetic worked out step by step, and the worked example laid bare so you can reproduce every number for yourself. Read it after the lecture, and keep it open during the lab.

Every figure quoted here comes from a single worked example, a credit-scoring model that we follow from its base rates through to the cost of trying to make it fair. The consistency is deliberate: the same numbers that fail one fairness test are the ones that make another test impossible to pass, and seeing that in one dataset is the fastest route to understanding why.

Setting up the tools (once)

You need Python and the same handful of libraries as before, plus one new one for the audit: Fairlearn. AIF360 is an optional heavier alternative used later in the lab.

terminal

python

```
python -m venv .venv && source .venv/bin/activate
pip install numpy pandas scikit-learn scipy matplotlib fairlearn
python week03-make_assets.py          # recreates every figure and printed number
```

What the libraries are for

NumPy does fast maths on arrays. **pandas** holds the table of applicants. **scikit-learn** supplies the model, the train/test split, and the confusion-matrix machinery from Week 2. **SciPy** adds statistics. **Matplotlib** draws the charts. **Fairlearn** is the new one: its **MetricFrame** computes any metric per group and reports the gaps in a single call.

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1 Foundations: why fairness is a modelling concern

The habit this module wants to break is the one that treats ethics as a final review step, a checklist applied to a finished model. That habit fails for a simple reason: by the time a model is finished, every decision that determines its fairness has already been made. The target you chose to predict, the loss you chose to minimise, the data you chose to learn from: each of these is a value judgement, and each is locked in long before anyone runs a fairness audit.

It helps to think of a model as an argument about the world rather than a neutral instrument. The argument runs: these features matter, this target is worth predicting, and this kind of mistake is more acceptable than that kind. None of those claims is neutral, and pretending otherwise is both an ethical failure and a technical one, because it hides your assumptions from yourself. The more useful question is not “is this model biased?” but “what does this model assume, and who benefits if the assumption holds?”

1.1 Accuracy is not a neutral objective

The phrase “just maximise accuracy” is often offered as the apolitical choice. It is not. Accuracy treats every error identically, whereas the real world attaches very different costs to different mistakes. A model that is ninety-five per cent accurate can still be catastrophic if the five per cent it fails are all the same people: the same neighbourhood, the same demographic, the same vulnerability. Accuracy averages that concentration away, and in doing so it quietly decides that a rare but concentrated harm is an acceptable price for a strong headline number.

The test to apply to any metric

Before accepting a metric, ask what it is willing to sacrifice, and whether you would state that sacrifice out loud to the people it affects. If the honest answer embarrasses you, you have found an ethical problem hiding inside a performance number.

1.2 Three ethical lenses

You do not need a philosophy degree to reason about fairness, but three lenses give you a vocabulary, and each responds differently to the same decision.

Consequentialist judges the model by outcomes overall. Its blind spot is distribution: it will happily benefit the many at the expense of a concentrated few.

Deontological judges the model by whether it follows fair rules and treats people as ends. It may reject a model that improves average outcomes, on grounds of principle.

Justice (Rawlsian) judges the model by the position of the worst-off. It may require you to give up aggregate performance to improve the lot of the least advantaged.

These lenses rarely agree, which leads to the single most practical instruction of the section: choose your lens before you tune the model, not after the numbers arrive. If you do not choose deliberately, the optimisation chooses for you, and it will always reach for the consequentialist average, because that is the easiest thing to optimise.

1.3 Whose fairness, and who decides

“Fair” is contested because the stakeholders want incompatible things. The applicant wants an equal chance; the lender wants precision about who repays; the regulator wants no disparate impact; and society, over the longer term, wants the historical gap to narrow rather than harden into the model. The hidden variable is power: the person who defines fairness (typically the data scientist) is rarely

the person the model decides about (the applicant). A useful professional habit is to notice, whenever a fairness definition is being chosen, who is in the room and who is not.

The section closes with a pivot that powers the rest of the booklet. The old question was “how accurate is it?”, a single number that is easy to report and easy to hide behind. The better question is “could you defend this decision to the person it was made about?” The moment you ask the second question, one number stops being enough, because you now need to know how the errors are distributed, not merely how many there are.

1.4 The reframe as a zero-sum game

The reframe deserves one more turn, because it is the question students ask most. “Is it biased?” treats bias as a defect inside the model, a fault you locate and delete, after which everyone is better off and nobody worse. That is *solitaire*: the payoff can only rise. But every assumption a model makes fits one group’s reality well and another’s badly. The group it fits well receives accurate, favourable treatment; the group it fits poorly absorbs the error. The instant you ask “who benefits if the assumption holds?”, you have stopped describing a property of the model and started describing an allocation between groups, and a benefit handed out implies one taken away.

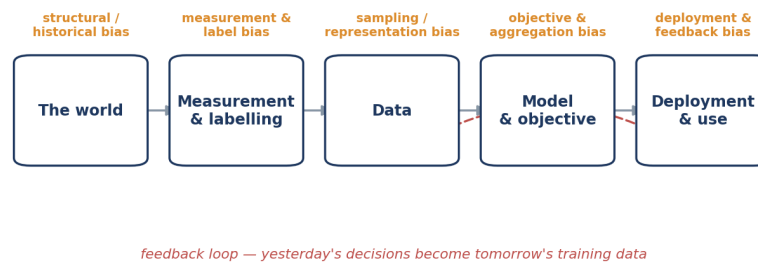
It is close to zero-sum because the model decides from a bounded budget: a fixed pool of approvals, one threshold, one objective. Under that constraint the fit is rivalrous. In the credit model, the assumption “length of credit history predicts repayment” holds for a group with long, well-recorded histories and misreads a group whose histories are thin through past exclusion; every approval the first group’s good fit secures at a given threshold is one the second group’s poor fit fails to secure, out of the same pool.

The honest caveat

It is not strictly zero-sum everywhere. Better features, better data or a genuinely better model can grow the pool and lift both groups at once, which is the cheap end of the accuracy-fairness frontier in Section 5. The zero-sum bite is sharpest at the frontier and when base rates genuinely differ, which is exactly where “just collect better data” runs out of road against societal bias.

2 Where bias enters the pipeline

Before measuring fairness, it pays to know where unfairness originates, because the instinct to blame “the algorithm” is usually misdirected and it lets the true cause escape scrutiny. Bias can enter at every stage between the world and the deployed decision, and a feedback loop feeds the last stage back into the first.



Bias enters at every stage. The dashed loop is the dangerous part: today's decisions become tomorrow's training data.

2.1 Three meanings of one word

The word “bias” carries three unrelated meanings, and conversations collapse when people use different ones without noticing.

Sense	Meaning	Typical remedy
Statistical	an estimator systematically away from the true value	more or better data; a better model
Social	a group receiving systematically worse outcomes	often not technical; sometimes structural
Cognitive	human judgement skewed by mental heuristics	process, review, diverse teams

The trap that catches good engineers

Reducing statistical bias can increase social bias. A more accurate model of an unjust world reproduces that injustice more faithfully. “We made it more accurate, therefore fairer” is a claim to distrust; accuracy and justice are different axes, and they sometimes point in opposite directions.

2.2 Data bias: the training set is a record, not a photograph

The training set records who was measured, who was counted as a success, and whose story was never captured. Four flavours recur:

Sampling bias some groups are over-represented or under-represented relative to the population the model will serve.

Historical bias past decisions are baked in and then relabelled as ground truth.

Measurement bias the proxy you could actually record differs from the quantity you meant to capture.

Labelling bias a human judgement, with its own slant, is frozen into the target variable.

The higher-order skill is to ask, for any dataset, which of these is most likely, and to design a test for it rather than assert its presence or absence.

2.3 The label is usually the real problem

A common and costly mistake is to scrutinise the features while trusting the label. Consider the credit model. The stated goal was to predict creditworthiness. The available label, however, is “did the bank approve them previously, and did they then repay”, a record shaped by who was offered credit at all. Training on that label does not teach the model who is creditworthy; it teaches the model who the previous system already favoured, and then presents that preference as an objective prediction.

Definition: proxy label

A target variable that stands in for the quantity you actually care about, because the true quantity is unmeasured or unmeasurable. When the proxy and the true target diverge, the model learns the divergence. Always audit the gap between the two before computing any metric, because that gap is precisely where bias hides.

2.4 Algorithmic and societal bias

Even with clean data, the algorithm adds slant. Objective functions reward exactly what you specified, including shortcuts you did not intend; aggregation optimises the average, which is a decision about whose errors may remain invisible; and regularisation pulls toward the majority pattern, smoothing minorities away because a minority resembles noise to a penalty that rewards simplicity. The optimiser is not malicious; it is faithful, and it executes an under-examined objective precisely.

Societal bias is the hardest layer, because it is not a collection mistake you can sample your way out of. Some gaps in the data are genuinely present, and they are present because the world that produced them was unequal. A model can be perfectly accurate about that world and still serve to entrench it. This raises a question with no neutral answer: should a model reflect the world as it is, or nudge toward the world as it should be? Reflecting reality faithfully encodes the inequality; correcting for it overrides the data with a values judgement. There is no view from nowhere, and declining to choose is itself a choice in favour of the status quo.

2.5 Unawareness fails, and feedback makes it worse

A tempting defence is to delete the sensitive attribute, so the model never sees it. This is called fairness through unawareness, and it does not work. In the worked example the model never receives group membership, yet from the model’s score alone the group can be recovered about seventy per cent of the time, against a base rate of forty-two per cent. Income, credit-history length and default counts are proxies for group in an unequal world, so the sensitive attribute is smeared across the other features (this is called redundant encoding), and removing one column changes very little.

Blindness is not fairness

“We do not use race or gender” is a comfortable defence and a weak one. Redundant encoding means the attribute is present in the model whether or not you named it. Fairness has to be measured on outcomes, not asserted from the absence of a column.

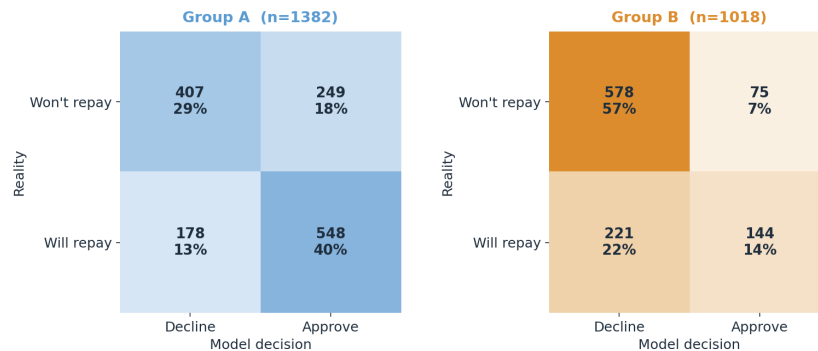
Feedback loops make the situation worse over time and are invisible to a single snapshot audit. If you deny loans to a group, you never observe whether they would have repaid, so next year’s training data holds even less signal about that group. The model grows more confident and more wrong, while appearing more accurate, because it is only ever tested on the people it already approved. The disparity is not merely measured by the system; it is manufactured by it.

3 Fairness, made precise

Until “fairness” is attached to a metric, it cannot be measured, argued about, or fixed. The mechanism for making it precise is one you already met in Week 2: the confusion matrix. The only new step is to compute one confusion matrix per group and then compare them.

3.1 The two group confusion matrices

For the credit model on the held-out test set, the two matrices are as follows. Positive means “approve”; the reality axis records whether the applicant would in fact repay.



The same model, one confusion matrix per group. Every fairness metric below is a comparison between these two tables.

Group	TN	FP	FN	TP
A (n = 1382)	407	249	178	548
B (n = 1018)	578	75	221	144

Here TN and TP are correct declines and correct approvals, FP is an approval that should have been a decline, and FN is a decline that should have been an approval. Every fairness number below is derived from these eight cells.

3.2 Group and individual fairness

Fairness metrics fall into two families that do not sit comfortably together.

Group fairness compares rates across groups, for example approval rates or error rates. It is straightforward to measure and is what regulators use, but it is blind to the individual.

Individual fairness requires that similar people receive similar treatment. It is intuitive but demanding, because you must define “similar”, and that definition reintroduces values through the back door.

The tension is real: enforcing group fairness can treat similar individuals differently across groups, while enforcing individual fairness can leave group rates far apart. You will have to choose, deliberately and defensibly.

3.3 Demographic parity

Also called statistical parity, demographic parity asks that the probability of a positive decision be equal across groups. It considers only the decision, never whether it was deserved.

$$P(\hat{Y} = 1 \mid A) = P(\hat{Y} = 1 \mid B)$$

the chance of a positive decision is identical whichever group you belong to

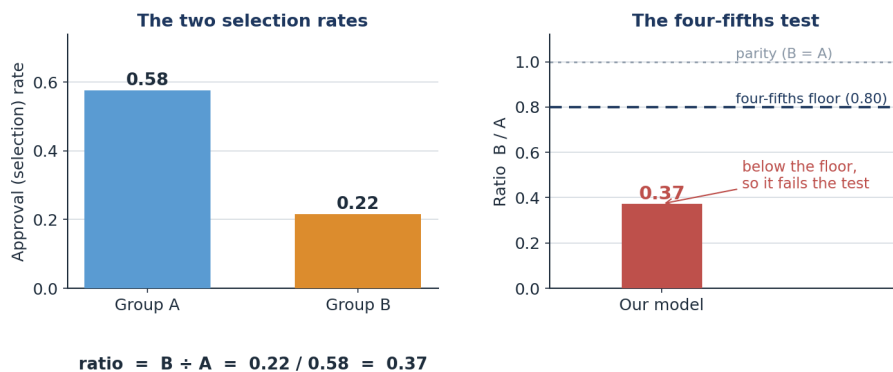
Demographic parity is simple, legible, and favoured by regulators. Its blind spot is that it ignores who actually repays: if the groups genuinely differ in outcome, parity can force the approval of worse-qualified applicants from one group in order to hit the quota. It is the right criterion when you believe the label itself is corrupted, that is, when differing base rates reflect past injustice rather than real difference. That belief is an argument you must make explicitly, not an assumption you may smuggle in.

3.4 Disparate impact and the four-fifths rule

Definition: disparate impact

The ratio of selection (positive) rates, disadvantaged group over advantaged group. A ratio below 0.80 is treated as prima facie evidence of disparate impact under the four-fifths rule, and in several jurisdictions it shifts the burden of justification onto the model's operator.

The four-fifths rule lives in ratio space, not rate space, and keeping the two apart avoids a common confusion. The selection rates are 0.577 and 0.215; their ratio, 0.37, is the number the rule tests against the 0.80 floor.



Left, the two selection rates and the division that yields the ratio. Right, the four-fifths test in ratio space: 0.37 against the 0.80 floor.

Worked calculation: selection rate and ratio

Selection rate is the fraction of a group that is approved, that is, (FP + TP) divided by n.

Group A: $(249 + 548) / 1382 = 797 / 1382 = 0.577$.

Group B: $(75 + 144) / 1018 = 219 / 1018 = 0.215$.

Disparate impact ratio: $0.215 / 0.577 = 0.373$. This is less than half of the 0.80 floor, so the model shows clear prima facie disparate impact.

3.5 Equalised odds and equal opportunity

Equalised odds asks that the errors not depend on group: equal true-positive rates and equal false-positive rates. A relaxation, equal opportunity, requires only equal true-positive rates.

$$\text{TPR}_A = \text{TPR}_B \quad \text{and} \quad \text{FPR}_A = \text{FPR}_B$$

among those who truly repay, equal approval rates; among those who truly do not, equal wrong-approval rates

Worked calculation: TPR and FPR

True-positive rate is $\text{TP} / (\text{TP} + \text{FN})$; false-positive rate is $\text{FP} / (\text{FP} + \text{TN})$.

Group A: $\text{TPR} = 548 / (548 + 178) = 548 / 726 = 0.755$; $\text{FPR} = 249 / (249 + 407) = 249 / 656 = 0.380$.

Group B: $\text{TPR} = 144 / (144 + 221) = 144 / 365 = 0.395$; $\text{FPR} = 75 / (75 + 578) = 75 / 653 = 0.115$.

The true-positive rates are 0.755 and 0.395. A qualified Group B applicant, one who would in fact repay, is therefore nearly half as likely to be approved as an equally qualified Group A applicant.

Equalised odds is often the more defensible criterion because it conditions on the truth: it does not demand equal approvals regardless of merit, only that being right or wrong does not depend on group membership.

3.6 Precision and calibration

Worked calculation: precision

Precision (positive predictive value) is $\text{TP} / (\text{TP} + \text{FP})$, the share of approvals that were correct.

Group A: $548 / (548 + 249) = 548 / 797 = 0.688$.

Group B: $144 / (144 + 75) = 144 / 219 = 0.658$.

Precision is nearly level across groups (0.688 against 0.658), even though selection and true-positive rates are far apart.

A closely related property is calibration: a score is calibrated if a predicted probability of 0.7 corresponds to an observed positive rate of about seventy per cent, and does so for both groups. The credit model is roughly calibrated, which produces a genuinely counter-intuitive situation: the model is simultaneously well-calibrated and precise across groups, yet wildly unequal on selection and true-positive rate. Both statements are true at once. If your working assumption was “calibrated implies fair”, this is where it breaks.

3.7 One model, four verdicts

Collecting the results, the same model receives four different verdicts depending on the criterion applied.

Criterion	Measured	Verdict
Demographic parity	0.577 vs 0.215	Fails
Disparate impact (floor 0.80)	ratio 0.373	Fails
Equal opportunity (TPR)	0.755 vs 0.395	Fails
Precision / calibration	0.688 vs 0.658	Passes

The lesson is that “is this model fair?” is unanswerable as posed. The same model is fair or unfair according to the criterion chosen, so the question that matters is “fair by which criterion, and can you justify choosing that one?” That is the question your ethical-impact assessment must answer.

4 Why you cannot satisfy every criterion

The four verdicts above are not a failure of effort; they reflect a mathematical result.

4.1 The impossibility result

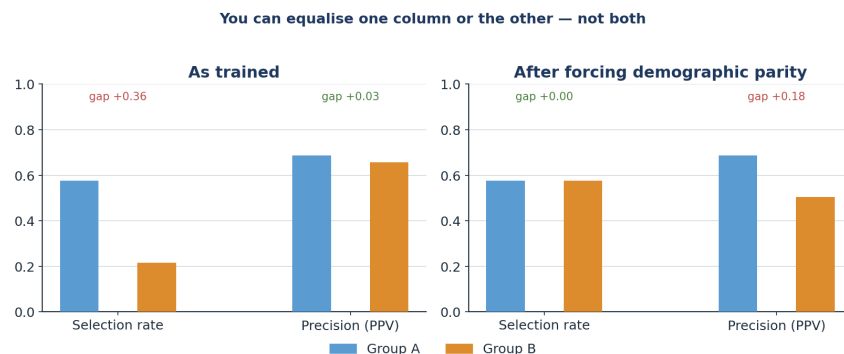
Impossibility (informal statement)

When two groups have different base rates, no classifier can simultaneously achieve equal calibration, equal false-positive rates, and equal false-negative rates, except in degenerate cases. You may satisfy some of these conditions, but not all at once. This was shown independently by Kleinberg, Mullainathan and Raghavan (2016) and by Chouldechova (2017).

The practical consequence is that any tool or vendor promising to make a model “fair” without qualification is offering a choice disguised as a solution: internally it selected which fairness condition to satisfy and dropped the others. Your responsibility is to know that this is happening and to make the choice yourself, in the open.

4.2 Watching the trade occur

The worked example makes the theorem concrete. As trained, the model has very different selection rates but roughly equal precision. Suppose we now force demographic parity by lowering Group B’s threshold until the two selection rates match at 0.577. The selection gap closes to zero, but Group B’s precision falls from 0.658 to roughly 0.50, while Group A’s precision holds near 0.688. We equalised selection and, in doing so, broke precision. We did not remove the unfairness; we moved it from one metric to another.



Force equal selection rates (right panel) and Group B’s precision collapses. The unfairness is transferred between columns, not removed.

Every fix is a transfer

The base-rate difference is what forces the trade. Because Group B repays at a lower observed rate, approving more of its applicants to match Group A’s selection rate necessarily lets in more who will not repay, which lowers precision for that group. The unfairness is conserved; the only real question is which transfer you can defend.

4.3 What algorithmic justice can and cannot deliver

What the apparatus can do is make trade-offs visible and quantified, force an explicit and auditable choice rather than an accidental one, and reveal disparities that a single accuracy number conceals. What it cannot do is invent a fairness that satisfies everyone (proven impossible), repair injustice that

lives in the data rather than the model, or substitute for the political decision about which harm is worse.

It follows that choosing one fairness criterion over another is not a technical preference but a stance on whether the base-rate difference is real or the residue of past injustice. That stance belongs to stakeholders, ethicists and the affected public, not to whoever wrote the training script. The professional's duty is to make the choice legible and honest, to price each option, and to refuse to let a default metric make a moral decision by accident.

5 The lab: audit, mitigate, cost

The lab turns the preceding analysis into a repeatable workflow applied to the Week 3 credit model.

5.1 Measuring group metrics

Measurement is short. Using Fairlearn, a `MetricFrame` computes any set of metrics per group and reports the gaps in one call.

audit.py

python

```
from fairlearn.metrics import MetricFrame, selection_rate
from sklearn.metrics import recall_score, precision_score

mf = MetricFrame(
    metrics={"selection": selection_rate,
            "tpr": recall_score,
            "ppv": precision_score},
    y_true=y_test, y_pred=y_hat,
    sensitive_features=group_test)

print(mf.by_group)      # per-group table
print(mf.difference())  # the gaps, in one call
```

mf.by_group / mf.difference()

	selection	tpr	ppv
A	0.577	0.755	0.688
B	0.215	0.395	0.658
difference	selection 0.362	tpr 0.360	ppv 0.030

The difficulty is not writing the code; it is reading the table and deciding what is tolerable. A 0.36 selection gap sitting beside a 0.03 precision gap is the whole ethical argument in three rows. AIF360 (from IBM) offers a broader alternative, with many more metrics and mitigation algorithms; prefer Fairlearn for a quick, defensible result, and reach for AIF360 when you want to compare several mitigations rather than apply one.

5.2 Three places to intervene

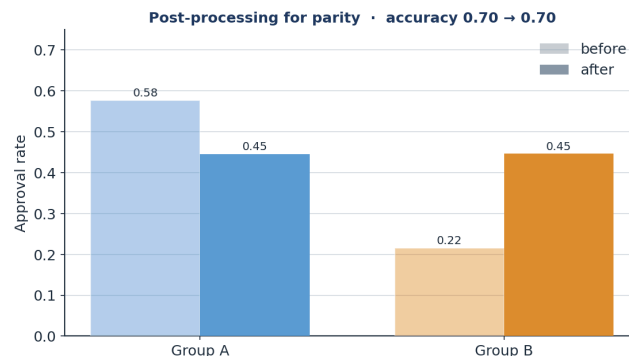
Mitigation happens at one of three stages, each with a distinct trade-off.

Stage	What you change	Trade-off
Pre-processing	the data: reweight, relabel, resample	touches the truth; can feel like cooking the books
In-processing	the training: fairness as a constraint	most principled; least portable across model types
Post-processing	the thresholds: per-group cut-offs	easy to add; means explicitly unequal treatment

None of the three is the correct one in general. Post-processing deserves particular caution: it means applying different decision thresholds to different groups, which can be defensible but reads, to an outsider, as using a different bar depending on group membership. Be ready to say so publicly.

5.3 Costing the fix, and naming who pays

The most important lesson of the lab overturns a lazy assumption. Sweeping the model toward parity traces a frontier of accuracy against the fairness gap; this frontier is a price list, not a free lunch, and you can read off what each increment of fairness costs.



Post-processing for parity. Aggregate accuracy barely moved, yet Group A's approvals fell from 0.58 to 0.45. The cost was redistribution, not accuracy.

Yet when post-processing was applied for parity in the worked example, aggregate accuracy barely moved, which might suggest that fairness was almost free. It was not. Group A's approval rate fell from 0.577 to about 0.45. The cost was not accuracy; it was redistribution, and the aggregate metric is exactly where the loser is hidden.

Ask who pays, not just how much

Fairness interventions are almost always redistributive. A stable aggregate metric does not mean nobody lost; it usually means the loss was moved between groups. Your assessment must name the transfer, not conceal it behind an unchanged headline number.

5.4 The ethical-impact assessment

The written deliverable is a one-page ethical-impact assessment, structured so that a sceptical reader cannot dismiss it. Five parts:

1. **Context:** what the model decides, about whom, and what is at stake for them.
2. **Findings:** the group metrics, stated plainly, worst gap first.
3. **Choice:** which fairness criterion you are holding the model to, and why that one.
4. **Cost:** what the mitigation spent, and who bore it.
5. **Residual risk:** what you did not fix, and what would make you refuse to deploy.

The final part is the one weak assessments omit and strong ones lead with. Being explicit about what you would not deploy is the clearest evidence that you understood the trade-offs rather than merely running the code.

Box-ticking audit	Strong audit
reports one fairness number	shows several metrics disagreeing
declares the model "fair"	names the criterion and defends it
hides the trade-off	quantifies what the fix costs
treats mitigation as a fix	states what it would refuse to deploy

The grade, and the professional respect, is in the defence rather than the model. The task is not to produce a fair model, which is provably impossible in general, but to produce a decision you can stand behind together with an honest account of its costs.

A Worked example at a glance

For reference during the lab, the full set of figures for the credit model, all derived from the two group confusion matrices in Section 3.

Quantity	Group A	Group B	Gap
Base rate (repays)	0.52	0.37	0.15
Selection rate	0.577	0.215	0.362
True-positive rate	0.755	0.395	0.360
False-positive rate	0.380	0.115	0.265
Precision (PPV)	0.688	0.658	0.030

The disparate impact ratio (Group B over Group A) is 0.373, well below the 0.80 four-fifths floor. Group can be recovered from the model's score alone about seventy per cent of the time (against a base rate of forty-two per cent), which demonstrates redundant encoding despite the sensitive attribute never being supplied to the model.

B Running everything yourself

One script, `week03-make_assets.py`, reproduces every number and figure in this booklet using a fixed random seed, so your results match exactly. From the project folder:

terminal**python**

```
python -m venv .venv && source .venv/bin/activate
pip install numpy pandas scikit-learn scipy matplotlib fairlearn
python week03-make_assets.py          # writes figs03/ and prints every result
```

It prints each labelled result (the base rates, the per-group confusion counts, the selection, TPR, FPR and precision values, the disparate impact ratio, the impossibility trade, and the cost of mitigation) and saves every chart into `figs03/`. The fastest way to build intuition is to change one thing at a time, for example the base-rate gap or the mitigation threshold, then run it again and watch the numbers and pictures respond.

C Glossary

A quick reference for the terms used in this booklet, each in one plain sentence.

Term	Plain meaning
Base rate	the underlying proportion of a group with the positive outcome, independent of any model
Calibration	predicted probabilities match observed frequencies, ideally within every group
Demographic parity	equal positive-decision rates across groups
Disparate impact	the ratio of selection rates; below 0.80 signals a legal concern under the four-fifths rule
Equalised odds	equal true-positive and false-positive rates across groups
Equal opportunity	the relaxation of equalised odds to true-positive rates only
Fairness through unawareness	the inadequate approach of removing the sensitive attribute from the inputs
Feedback loop	a model's own decisions shape the data on which it is later trained
Proxy label	a measurable stand-in for the true target, which may encode the very bias you meant to avoid
Redundant encoding	a sensitive attribute present through correlated features, even when the attribute is removed

Going further. J. Kleinberg, S. Mullainathan and M. Raghavan, “Inherent Trade-Offs in the Fair Determination of Risk Scores”, *ITCS*, 2016 (the impossibility result; read this one first). · A. Chouldechova, “Fair Prediction with Disparate Impact”, *Big Data* 5(2), 2017. · M. Hardt, E. Price and N. Srebro, “Equality of Opportunity in Supervised Learning”, *NeurIPS*, 2016. · S. Barocas and A. D. Selbst, “Big Data’s Disparate Impact”, *California Law Review* 104, 2016. · J. Buolamwini and T. Gebru, “Gender Shades”, *PMLR* 81, 2018. · S. Barocas, M. Hardt and A. Narayanan, *Fairness and Machine Learning* (fairmlbook.org). · C. O’Neil, *Weapons of Math Destruction*, 2016. · The Fairlearn user guide (fairlearn.org) and AIF360 documentation (aif360.res.ibm.com).